

Task 1:

Define a function, `cube_num(num)` that takes a number as input, cubes it, and then returns it. You can assume the user always provides a number as the argument.

Task 2:

Define a function, `is_even(num)` that takes a number as input, and returns whether it's even. You can assume the user always provides a number as the argument.

Task 3:

Define a function, `quadratic_roots(a, b, c)` that solves for and returns the roots of a polynomial in the format $ax^2 + bx + c$. Recall that the formula for doing so is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Make sure your function returns *both* roots! We haven't seen how to do this yet, but try some different approaches and see if they work.

Task 4:

Last week, we wrote some code to determine if a number is prime or not. Take that code, bottle it up in a function `is_prime`, which takes an `int` and returns its primality as a boolean (`True/False`).

Then, *outside* of the function, write another loop through all of the numbers from 1 to 100. For each number, call your `is_prime` function to see if the number is prime. If it is, print it out. Otherwise, do nothing. The end result of running this is you should see all prime numbers between 1 and 100 printed.